P P SAVANI UNIVERSITY

Seventh Semester of BCA/B.Sc.IT Examination

October 2026

SSCS4010 DevOps and Agile Foundation

16.10.2026, Friday **Time: 12:30 p.m. To 03:00 p.m.** **Maximum Marks: 60**

Instructions:

1. The question paper comprises of two sections.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator is allowed.

SECTION – I

Q - 1	**MCQ/Short Question/Fill in the Blanks (Any Five)**	**[05]**	**CO**	**BTL**
(i)	Which of the following is NOT one of the four values of the Agile Manifesto? (a) Individuals and interactions over processes and tools (b) Working software over comprehensive documentation (c) Following a plan over responding to change (d) Customer collaboration over contract negotiation		**1**	**1**
(ii)	In Scrum, the person responsible for maximizing product value and managing the Product Backlog is the _____.		**2**	**1**
(iii)	Which Git command is used to copy an existing remote repository to the local machine? (a) git init (b) git clone (c) git push (d) git commit		**1**	**2**
(iv)	The Git command _____ moves changes from the working directory to the staging area.		**2**	**1**
(v)	What does WIP stand for in Kanban?		**3**	**2**
(vi)	A declarative Jenkins pipeline is written in a file named: (a) pom.xml (b) Dockerfile (c) Jenkinsfile (d) ci.yml		**1**	**1**
(vii)	In GitHub Actions, workflow files are stored in the _____ directory of a repository.		**3**	**2**
Q - 2	**Attempt any two.**	**[10]**		
(i)	State the four values of the Agile Manifesto and list any four Agile principles.		**1**	**2**
(ii)	Explain the three roles and three artifacts of the Scrum framework.		**3**	**2**
(iii)	List and state the purpose of the following Git commands: init, clone, status, commit and push.		**1**	**3**
(iv)	What is Continuous Integration? List any four key CI practices.		**3**	**4**
Q - 3	**Attempt any two.**	**[10]**		
(i)	Briefly explain the "Three Ways" of DevOps with one example practice for each.		**2**	**1**
(ii)	(a) Write two user stories for an online examination portal in the "As a... I want... so that..." format. (b) A team completed 28, 32, 30 and 34 story points in its last four sprints. Calculate its average velocity.		**1**	**5**
(iii)	Explain the three areas of Git (working directory, staging area and repository) with a neat diagram.		**2**	**2**
Q - 4	**Attempt any one.**	**[05]**		
(i)	Draw and explain a typical CI workflow from code commit to build artifact.		**2**	**2**
(ii)	Differentiate between Scrum and Kanban (any five points).		**2**	**5**

SECTION II

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Q - 1	**MCQ/Short Question/Fill in the Blanks (Any Five)**	**[05]**		
(i)	In which practice is every change that passes automated tests released to production automatically, without a manual approval step? (a) Continuous Integration (b) Continuous Delivery (c) Continuous Deployment (d) Continuous Testing		**1**	**1**
(ii)	A feature flag used to instantly switch off a misbehaving feature in production is called a ___off___ switch.		**2**	**2**
(iii)	Which Terraform command shows the proposed changes before they are made? (a) terraform init (b) terraform plan (c) terraform apply (d) terraform destroy		**1**	**1**
(iv)	Why is Ansible called "agentless"?		**3**	**2**
(v)	The default Service type in Kubernetes is: (a) NodePort (b) LoadBalancer (c) ClusterIP (d) Ingress		**2**	**2**
(vi)	In Docker, a running instance of an image is called a _____.		**4**	**1**
(vii)	Name the three pillars of observability.		**1**	**1**
Q - 2	**Attempt any two.**	**[10]**		
(i)	Define Continuous Integration, Continuous Delivery and Continuous Deployment. Draw a simple deployment pipeline.		**1**	**5**
(ii)	What are feature flags? Explain any four types of feature flags with one example each.		**2**	**2**
(iii)	Differentiate between containers and virtual machines (any five points).		**1**	**5**
(iv)	Explain the Terraform commands init, plan, apply and destroy. State the role of the state file.		**2**	**2**
Q - 3	**Attempt any two.**	**[10]**		
(i)	Explain blue/green deployment with a neat diagram. State its advantages.		**1**	**1**
(ii)	Explain the following principles of Infrastructure as Code: declarative vs. imperative, idempotence and version control.		**2**	**2**
(iii)	In one month a team made 40 production deployments, of which 6 caused incidents. Three production outages were restored in 30, 90 and 60 minutes. Calculate the Change Failure Rate and the Mean Time to Restore (MTTR).		**1**	**6**
Q - 4	**Attempt any one.**	**[05]**		
(i)	Explain Docker architecture (client, daemon, images, containers and registry) with a neat diagram.		**2**	**5**
(ii)	Explain Pod, ReplicaSet, Deployment and Service in Kubernetes.		**1**	**4**

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CO : Course Outcome Number BTL : Blooms Taxonomy Level

Level of Bloom's Revised Taxonomy in Assessment

1: Remember	2: Understand	3: Apply
4: Analyze	5: Evaluate	6: Create